Daniel Zhang

2nd Year PhD Student · Statistics

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Education

University of Waterloo *Waterloo, ON*

PhD Candidate - Statistics *Jan, 2025 - Present*

- Research Interest: Deep Learning Theory, Bayesian Deep Learning, Variational Inference
- Supervisor: Prof. Ali Ghodsi
- Grade: 91/100

University of Waterloo *Waterloo, ON*

Masters of Mathematics - Statistics *Sept, 2023 - Aug, 2024*

- Research focus: Natural Language Processing, Generative AI, Generalized Additive Models
- Supervisor: Prof. Ali Ghodsi
- Grade: 91/100

University of Waterloo *Waterloo, ON*

Honours Bachelors of Mathematics - Statistics *Sept, 2018 - Apr, 2023*

- Minor in Computing, Co-operative Program
- Undergraduate research advisors: Prof. Alexander Wong, Dr. Jason Deglint, Vikram Voleti

Research Experience

University of Waterloo - Data Analytics Lab *Waterloo, ON*

Supervisors: Prof. Ali Ghodsi *Sept 2023 - Aug, 2024*

- **Project:** ***GAM SymbolicGPT: A Generalized Additive Model Approach to SymbolicGPT***
- Explored symbolic regression to generate interpretable mathematical expressions capturing functional relationships between inputs and outputs
- Analyzed the limitations of deep learning-based symbolic regression methods in high-dimensional settings, highlighting the impact of exponentially large expression search spaces
- Proposed and implemented *GAM SymbolicGPT*, a novel extension of SymbolicGPT that incorporates additive structure via a backfitting-inspired optimization scheme
- Designed and executed experiments demonstrating the advantages and constraints of the method in synthetic and real-world high-dimensional regression tasks

University of Waterloo - Vision and Image Processing Lab *Waterloo, ON*

Supervisors: Prof. Alexander Wong, Dr. Jason Deglint, Vikram Voleti *Sept - Dec, 2022*

- **Projects:** ***Plankton-FL***, ***Git Re-Basin Federated Learning***
- Optimized Federated Learning performance through systematic grid search over learning rates and client participation ratios, improving convergence and model accuracy
- Applied the *Git Re-Basin* approach to align model parameter spaces across non-overlapping datasets, enabling effective phytoplankton classification from disjoint data sources
- Led the project to publication as first author; research on Federated Learning for cross-source plankton classification was accepted to CVIS 2022

Blue Lion Labs *Waterloo, ON*

Supervisors: Dr. Jason Deglint, Vikram Voleti *July - Aug, 2022*

- Investigated cutting-edge variants of Federated Learning and contributed novel architectural and training strategies to the Research and Development team
- Designed and implemented a Federated Learning framework using convolutional neural networks to classify phytoplankton species from heterogeneous, distributed datasets

National Research Council Canada *Toronto, ON*

SUPERVISORS: DR. MOHAMMAD SAJJAD GHAEMI, DR. JAMES H.K. OOI *Jan - Apr, 2021*

- **Project: *Transfer Learning for COVID-19 Patient Severity Classification***
- Explored diverse applications of Transfer Learning and presented actionable insights to guide modeling strategy for clinical outcome prediction.
- Developed a two-stage Transfer Learning framework: pre-trained an MLP on Influenza patient data, then fine-tuned on COVID-19 clinical records for severity classification.
- Benchmarked the Transfer Learning model against a baseline MLP trained from scratch, achieving a 25% improvement in AUROC for COVID-19 severity prediction.
- Engineered clinical feature representations and addressed class imbalance using Synthetic Minority Oversampling Technique (SMOTE) to enhance model generalization.

Publications

PUBLISHED

Zhang, D, Voleti, V, Wong, A, Deglint, J. 2022. Plankton-FL: Exploration of Federated Learning for Privacy-Preserving Training of Deep Neural Networks for Phytoplankton Classification. Proceedings of the 8th Annual Conference on Computational Vision and Intelligent Systems

Awards, Fellowships, & Grants

2026	**Statistics & Actuarial Science Chair's Graduate Award**, University of Waterloo
2024, 2025	**Graduate Research Studentship**, University of Waterloo
2022	**Best AI Paper Award**, Waterloo AI Institute
2018	**President's Scholarship**, University of Waterloo

Projects

BNN-μP: A μP Scaling Framework for Variational Inference

STAT946 PROJECT

- Proposed **BNN-μP**, a principled extension of Maximal Update Parametrization (μP) to Bayesian Neural Networks, addressing scaling pathologies in variational inference for wide models.
- Derived width-aware scaling laws for **learning rates, variational parameter updates, and KL regularization**, preventing posterior collapse and KL explosion in the infinite-width limit.
- Reformulated the ELBO objective with an **effective regularization normalization** and KL warm-up schedule to stabilize Bayesian training dynamics under μP scaling.
- Demonstrated **zero-shot hyperparameter transfer** across network widths (256→2048) on CIFAR-10, eliminating the need for width-specific retuning.
- Implemented **joint Bayesian hyperparameter optimization** with scaling-aware parameterization, successfully identifying width-invariant optimal configurations on a small proxy model.

Bayesian NoProp: Enabling Uncertainty Quantification for Back-Propagation-Free Neural Networks

STAT900 PROJECT

- Goal: Extended the NoProp framework by incorporating Bayesian inference to enable principled uncertainty quantification while maintaining its back-propagation-free training paradigm.
- Developed Bayesian extensions of the *NoProp-DT* framework by introducing probabilistic linear and convolutional layers, enabling principled modeling of epistemic and aleatoric uncertainty.
- Implemented Bayesian-MLP and Bayesian-End-to-End variants with KL-regularized variational inference, Monte Carlo sampling, and annealed KL schedules to stabilize training.
- Conducted extensive experiments on CIFAR-10/100 across multiple label embedding strategies, achieving up to 5% higher test accuracy and significantly better calibration compared to frequentist NoProp.
- Designed and executed uncertainty quantification studies under distribution shift, demonstrating meaningful increases in both epistemic and aleatoric uncertainty.

Trust Weighted Aggregation of Stochastic Generations for Uncertainty-Aware Inference in Large Language Models

STAT900 Project

- Goal: Developed a principled method for uncertainty quantification in LLMs by aggregating multiple stochastic generations using trust-weighted model averaging.
- Formulated LLM decoding as sampling from an implicit posterior over reasoning paths, allowing predictive entropy to serve as a measure of epistemic uncertainty.
- Introduced trust scores based on accuracy and embedding variance to weight model outputs, improving interpretability and robustness of aggregated predictions.
- Evaluated across deterministic and open-ended prompts using metrics like entropy, expected calibration error, and Jensen-Shannon divergence to quantify uncertainty, calibration, and semantic robustness.
- Demonstrated that trust-weighted aggregation enhances interpretability and model reliability without requiring access to model internals.

Combining Deep Learning and Bayesian Inference for Generalized Extreme Value (GEV) Parameter Estimation and Uncertainty Quantification

STAT946 Final Project

- Developed scalable deep learning approaches for estimating GEV distribution parameters and quantifying uncertainty in spatial extreme value modeling.
- Implemented feedforward and Bayesian neural networks with approximate inference methods including MC Dropout, Laplace Approximation, Bayes-by-Backprop, MFVI, and SG-MCMC.
- Benchmarked neural approaches against traditional statistical methods such as Gaussian Process Regression on small- and large-scale spatially simulated datasets.
- Demonstrated that Bayesian deep learning models like SG-MCMC yield better-calibrated uncertainty than variational methods, albeit at higher computational cost.
- Showed that neural models scale more efficiently than classical methods, achieving fast inference while recovering complex spatial structure.

Can LLMs Distinguish AI-Driven Review Bombing and Boosting of Businesses?

STAT940 Final Project

- Classified between human-written and AI-generated restaurant reviews
- To address the significant class imbalance, multiple open-source APIs, including GPT, Llama, and Cohere, were utilized to generate synthentic data.
- Developed two innovative optimizers: Adam with Gradient Clipping, a variant of the Adam optimizer that clips gradients to avoid gradient explosion, and ElasticAdam, an adaptation of Adam that incorporates both L1 and L2 penalties.
- Fine-tuned a basic BERT model to attain an accuracy of 71% and an F1 score of 61%, and then conducted empirical comparisons with a BERT model that was optimized using our innovative optimizers.
- Enhanced the baseline model's performance, achieving a 7% increase in accuracy and an 18% improvement in the F1 score, utilizing the Adam with Gradient Clipping optimizer.

Teaching Experience

Teaching Assistant, University of Waterloo — Spring '26

STAT444/844 - Statistical Learning: Advanced Regression

Teaching Assistant, University of Waterloo — Winter '26

STAT946 - Case Studies in Data Science

Teaching Assistant, University of Waterloo — Fall '25

STAT845 - Statistical Concepts for Data Science

Teaching Assistant, University of Waterloo — Winter '25

STAT441/841 - Statistical Learning: Classification

Teaching Assistant, University of Waterloo — Winter '25, Spring '25, '26
STAT341 - COMPUTATIONAL STATISTICS AND DATA ANALYSIS

Teaching Assistant, University of Waterloo — Spring '24
STAT443 - FORECASTING

Head Teaching Assistant, University of Waterloo — Winter '24, '26
STAT442/842 - DATA VISUALIZATION

Head Teaching Assistant, University of Waterloo — Winter '24
STAT940 - DEEP LEARNING

Teaching Assistant, University of Waterloo — Fall '23,'25
STAT442/842 - DATA VISUALIZATION

Teaching Assistant, University of Waterloo — Fall '23, Spring '24, Spring'25
STAT231 - STATISTICS

Industry Experience

Intact Financial Corporation — *Toronto, ON*
DATA SCIENTIST I — *Jun, 2023 - Dec, 2024*

- Contributed to the User-Based Insurance (UBI) team with a focus on the Eco & Fuel and Driving Model initiatives, supporting the analysis of over 500,000 telematics trips to improve driver risk assessment
- Led validation efforts for the Eco & Fuel feature across iOS and Android platforms, resolving 10+ bugs and edge cases prior to launch, enabling a production release used by over 100,000 users
- Collaborated with cross functional teams to implement core logic for driving event detection and score computation, covering over 10 event types including hard braking and rapid acceleration, for the new driving model deployed to live UBI scoring
- Conducted feature importance and redundancy analysis on 100+ telematics features, reducing the active feature set by 25% and improving the Gini coefficient of the model by 2%.

Intact Financial Corporation — *Toronto, ON*
DATA SCIENCE INTERN — *Sept - Dec, 2022*

- Supported the User-Based Insurance (UBI) team on the Commercial Lines Trucking initiative, contributing to model development for a portfolio of over 50,000 commercial vehicles
- Designed and deployed scalable data pipelines processing 2M+ daily telematics records, normalizing over 150 high-dimensional features and linking them to historical claims data for modeling
- Performed correlation and significance analysis on 120+ aggregated features, identifying the top 15 predictive variables linked to claims severity and frequency
- Reviewed and refactored 10,000+ lines of code across peer contributions, increasing maintainability and reducing runtime errors by over 30% through rigorous review practices

TD Bank — *Toronto, ON*
DATA SCIENCE AND ADVANCED ANALYTICS INTERN — *May - Aug, 2022*

- Contributed to the TD Wealth Data Analytics team, supporting data initiatives that impacted over 200,000 Direct Investing clients and streamlined internal operations across business units.
- Automated 5+ legacy business workflows using SAS and PowerShell, reducing manual processing time by 70% and improving data delivery reliability.
- Built and maintained SQL and PySpark pipelines that ingested and processed 10M+ records daily, enabling near real-time updates to client dashboards and reporting systems.
- Contributed to the migration of 10+ ETL workflows to Azure, leveraging PySpark and Hadoop to improve processing speed by 3x and establish a scalable cloud-native data infrastructure.

Intact Financial Corporation *Toronto, ON*

DATA SCIENCE INTERN *Sept - Dec, 2021*

- Supported the Rating-Revolution (RR) team on the Tooling Project, contributing to tooling enhancements used by 5+ actuarial teams to accelerate insurance pricing model development.
- Refactored PySpark pipelines handling over 5M records, reducing end-to-end data processing time by 60% and enabling faster model iteration cycles.
- Integrated a modular pre-processing layer into the Pandas workflow, allowing actuaries to inject custom features and parameters with over 75% reduction in manual data prep effort.